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Ministry of Education



RTB | RWANDA
TVET BOARD

CURRICULUM STRUCTURE

RQF LEVEL

4

TVET CERTIFICATE IV In SOFTWARE DEVELOPMENT

ICTSWD4002

Kigali, September 2023

ICTSWD4002-TVET CERTIFICATE IV

Software Development

RQF Level 4 CURRICULUM

Funded by:



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List of Abbreviations

ICT	Information and Communication Technology
IT	Information Technology
NISR	National Institute of Statistics of Rwanda
NST1	National Strategic Transformation 1
RQF	Rwanda Qualification Framework
RTB	Rwanda TVET Board
SDG	Sustainable Development Goals
TQUM	TVET Quality Management
TVET	Technical Education and Vocational Training

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FOREWORD

The global rapid changes in the labour market tends to call for appropriate training and skills development through outcome-based training approaches. Skills development and employment promotion are central to Rwanda's transformative Vision 2050, aiming to secure high standards of living for all Rwandans. In a bid to transform Rwanda into a knowledge based economy, the National Strategic Transformation 1 (NST1) calls socio-economic transformation through TVET skills development. The Rwanda TVET Board (RTB) was established to promote quality education in technical and vocational education and training from level one (1) to five (5) aimed at fast tracking socio-economic development of the country. Designing and distributing curricula, teaching materials, trainer's guides, methodologies and establish training methods for technical and vocational education and training from level one (1) to five (5); is among other RTB's responsibilities

The existing curricula were limited and narrowed in terms of acquired skills and knowledge and were not meeting the requirements of the current labour market at both national and regional level. In addition, there were. Barriers in vertical mobility and pathways in TVET which resulted in negative TVET perception. Furthermore, there were barriers to admission of TVET graduates of certain programs into higher learning institutions.

The TVET modernization process has begun with a clear picture of the programs focusing on sectors with the high employment potential like Software development among others. In this respect, Rwanda TVET Board, is honoured to avail the curriculum of Software development which serves as the official document and respond to the above mentioned concerns.

With the help of the training providers, trainers, parents whose role is central to the success of this curriculum, the trainees will gain appropriate hand on skills which will make a difference not only to their own lives but also to the success of Rwanda's economy.

I wish to sincerely extend my appreciation to the people who contributed towards the development of this document.

Dipl.-Ing. Paul UMUKUNZI

Director General/ RTB

Rationale of the Qualification

A Backend Developer contributes to Rwanda's digital transformation and Internet based services as stated in ICT Sector Strategic Plan (2018-2024), several achievements have been registered by the sector in areas related to ICT infrastructure, websites development, business and investments, system administration, global partnerships, among others.

Over the last decade and half, the Gross Domestic Product of the ICT sector in Rwanda has been growing at a median rate of 15.3% (NISR,2016). Internet Penetration in Rwanda is increasing steadily. For instance, internet penetration increased from 7% in 2011 to 39.76% mid 2017. While mobile increased from 639,673 to 9.7 million over the period 2010 -2017

In order to attain long term developmental goals of Rwanda and Africa, such as the Sustainable Development Goals (SDG), Rwanda's Vision 2050 and African Union's common goals under the Agenda 2063, it is essential to maximize the power of ICT as a true enabler of socio-economic development. The ICT sector remains one of the primary targets to boost country's economy, which will be achieved through:

- Positioning Rwanda as a world class and high end ICT hub
- Job creation that reduces the poverty
- Contribution to GDP growth
- Facilitation of cost-effective public and private services
- Impact of ICT-related capital investments on overall capital deepening
- Increase of labor productivity & significant contribution to value-added by ICT jobs
 - Digital Literacy for all to promote integration of isolated communities into the national economy & Increase the access of larger markets coverage
 - Developing the ICT industry and build the capacity of the private sector to provide high levels of service delivery.

1. GENERAL INTRODUCTION

The curriculum presents a coherent and significant set of competencies to acquire to perform the occupation of a backend developer. It is designed with an approach that takes into account the training needs, the work situation, as well as the goals and the means to implement training.

The modules of the curriculum include a description of the expected results at the end of training. They have a direct influence on the choice of theoretical and practical learning activities. The competencies are the targets of training: the acquisition of each is required for certification.

The curriculum is the reference to carry out the assessment of learning. Assessment tools of learning are developed on the basis of this document.

The curriculum consists of three parts. The first part is of general interest and shows the nature and goals of a program and the key concepts and definitions used in the document. The second part presents the qualification, its level in the qualification framework, its purpose, its rationale and the list of modules it comprises. The third part deals with the training package. It includes the competencies chart, the sequencing of module learning, the description of each module and the course structure.

The pages describing the modules are the heart of a curriculum. They present the title of the module, the length of training, the amount of credits, the context in which the competency is performed, the prerequisite competencies, the learning outcomes and the performance criteria.

In each module, a course structure is provided. The course structure describes the learning outcomes (knowledge, skills and attitude) and the learning contents related to each learning outcome. Also, the learning activities and resources for learning are suggested.

Finally, the assessment specifications and guidelines are included in each module.

2. QUALIFICATION DETAILS

2.1. Description

Title:	TVET Certificate IV in Software Development
Level:	RQF Level 4
Credits:	120
Sector:	ICT and Multimedia
Trade:	Software Development
Issue date:	September, 2023

2.2 Graduate Profile

This qualification provides the skills, knowledge and attitudes for a learner to be competent in tasks and activities that require the application of practical skills in a defined context. Work would be undertaken in various Software Development sectors/industries where Software Development includes Database development, Apply Algorithm and Data Structure Fundamentals using JavaScript, Design Backend system, Apply PHP programming, develop a backend application using Node Js, Perform basic window server administration. Learners may work with some autonomy or in a team but usually under close supervision.

At the end of this qualification, qualified learners will be able to:

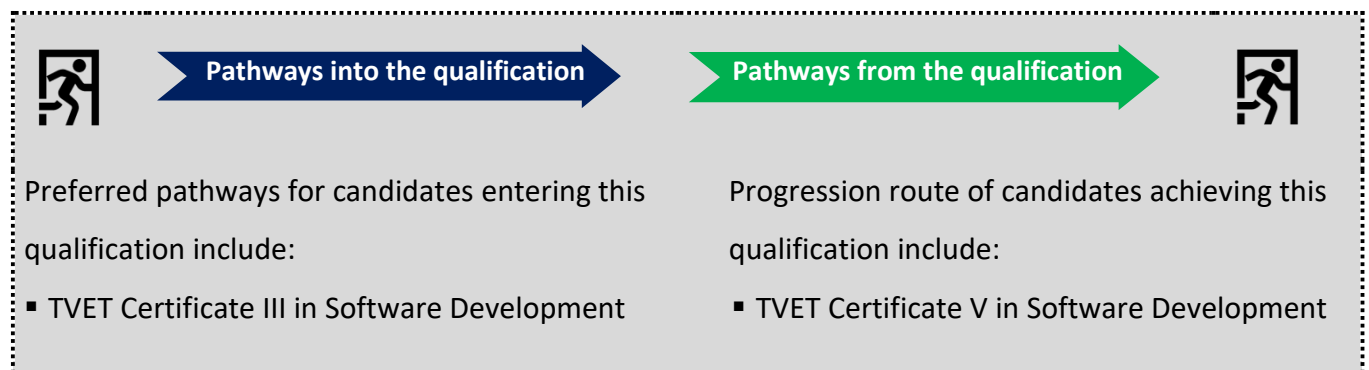
1. Develop a business plan
2. Apply Computer Skills
3. Promote the Culture of Peace
4. Use intermediate English at the workplace
5. Gukoresha Ikinyarwanda Cy'Umunyamwuga

6. Pratiquer les activités de communication en Français dans le métier
7. Use Applied Mathematics
8. Use Applied Physics
9. Develop Database
10. Apply Data Structure and Algorithm Fundamentals using JavaScript
11. Design Backend System
12. Apply PHP Programming
13. Develop a Backend Application using Node.js
14. Perform Basic Windows Server Administration
15. Apply Basics of Networking
16. Integrate the workplace

2.3 Minimum entry requirements

The minimum entry requirement to this course is to be mentally fit with an interest in software development and related areas

2.4 Information about pathways



2.5 Job related information

This qualification prepares individuals to integrate in small and medium ICT companies, he/she can also work in other public and/or private sectors that require ICT related services with the professionalization of Software Development, while ensuring that safety, security and environmental regulations are respected.



Possible jobs related to this qualification

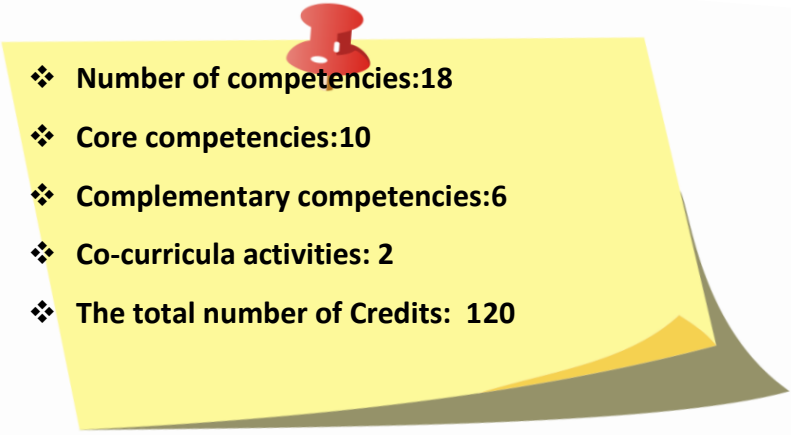
- Junior Webmaster
- Junior Back End Developer
- Junior Web Application Developer
- SQL Developer
- PHP developer
- Laravel developer

2.6 Information about competences

No	Code	Cross-cutting competencies	Credit
1.	CCMBC402	Develop a business plan	3
2.	CCMCL402	Apply computer skills	3
3.	CCMCT401	Promote the Culture of Peace	3
4.	CCMEN402	Use intermediate English at workplace	3
5.	CCMKN402	Gukoresha Ikinyarwanda kiboneye	3
6.	CCMFT402	Exprimer des opinions en français élémentaire	3
Total			18

No	Co-Curricula activities	Credit
1	Sports/clubs	1
2	Self-study/ library research	1
Total		2

	No	Code	Core competencies	Credit
GENERAL	1.	GENFA402	Apply Fundamental Mathematical Analysis	6
	2.	GENMP402	Apply Mechanics and properties of matter	4
	3.	GENBN401	Perform Basics of Networking	7
SPECIFIC	4.	SWDDD401	Develop Database	10
	5.	SWDDA401	Apply Data Structure and Algorithm Fundamentals using JavaScript	13
	6.	SWDBS401	Design Backend System	8
	7.	SWDPP401	Apply PHP Programming	13
	8.	SWDBA401	Develop a Backend Application using Node.js	10
	9.	SWDWS401	Perform Basic Windows Server Administration	9
	10.	SWDIA401	Integrate the workplace	20
Total				100

- 
- ❖ Number of competencies:18
 - ❖ Core competencies:10
 - ❖ Complementary competencies:6
 - ❖ Co-curricula activities: 2
 - ❖ The total number of Credits: 120

2.7 Allocation of Learning Hours

NO	Module name	Learning outcome	Theoretical hours	Practical hours	Total hours
1	Entrepreneurship	LO1. Identify elements of business plan	7	3	10
		LO2. Write a business plan	3	7	10
		LO3. Establish business contingency plan	3	4	7
		LO4. Present a business plan	1	2	3
Total hours module 1			14	16	30
2	ICT	LO1. Describe the operating system	5	10	15
		LO2. Customize the computer features	4	6	10
		LO3. Protect computer system	2	3	5
Total hours module 2			11	19	30
3	Citizenship	LO1. Make a Comparative study of Genocides.	3	7	10
		LO2. Protect human rights	3	7	10
		L.O3. Promote social cohesion	3	7	10
Total hours module 3			9	21	30
4	English	L.O1. Write factual, descriptive, and explanatory correspondence text.	4	5	9
		L.O2. Apply a range of listening strategies and activities to understand predictable messages	3	4	7
		L.O3. Discussion, supporting or refute ideas on general and trade-related topics	2	5	7

		L.O4. Read medium texts on general and trade-related topics	2	5	7
Total hours module 4			11	19	30
5	Ikinyarwanda	L.O1. Gukoresha ubuvanganzo gakondo bufatiye ku mwuga ashyikirana n'abandi , gutandukanya ubwoko bw'inshinga no gukoresha ikomorazina n'ikomoranshinga .	6	4	10
		L.O2. Gukoresha Ikinyarwanda uwiga agaragaza ibyiza by'ikoranabuhanga mu iterambere ry'umwuga no gusesengura ibinyazina binyuranye.	2	3	5
		L.O3. Gukoresha Ikinyarwanda uwiga agaragaza ububi bw'ibiyobyabwenge mu rubyiruko, kugaragaza amategeko y'igenamajwi yerekeye n'ingombajwi mu izina mbonera no kunoza imyandikire y'inyuguti nkuru mu nteruro.	2	3	5
		L.O4. Gukoresha Ikinyarwanda uwiga agaragaza uburyo bunyuranye bwo gufata neza ibidukikije no gukoresha indangahantu n'ibyungo.	2	3	5
		L.O5. Gukoresha Ikinyarwanda uwiga agaragaza akamaro k'ubutabazi bw'ibanze no gukoresha impakanyu	2	3	5

		n'indango z'inshinga.			
Total hours module 5			14	16	30
6	Français	L.O1. Préciser des informations,	6	5	11
		L.O2. Décrire des personnes et des lieux et exprimer ses goûts	3	5	8
		L.O3. Indiquer l'itinéraire.	5	6	11
Total hours module 6			14	16	30
7	Applied Mathematics	LO1: Analyse algebraic functions	5	10	15
		LO2: Apply fundamentals of differentiation	8	12	20
		LO3: Apply exponential functions	5	10	15
		LO4: Apply logarithmic functions	3	7	10
Total hours module 7			21	39	60
8	Applied Physics	LO1: Describe laws of motion and their applications	2	2	4
		LO2: Apply static equilibrium and elasticity	3	5	8
		LO3: Analyse fluid mechanics	2	4	6
		LO4: Apply thermodynamics	3	6	9
		LO5: Examine effects of electric current flow in DC electric circuit	3	4	7
		LO6: Apply geometric instruments	2	4	6
Total hours module 8			15	25	40
9	Basics of Networking	Establish network media connectivity	6	14	20
		Perform Basic Network	10	25	35

		Configuration			
		Maintain Network system	5	10	15
Total hours module 9			21	49	70
10	Database Development	Analyse Database	10	10	20
		Design Database	5	20	25
		Implement Database	10	26	36
		Secure Database	5	14	19
Total hours module 10			30	70	100
11	Data Structure and Algorithm Fundamentals	Apply Algorithm Fundamentals	15	15	30
		Apply Data Structure	10	35	45
		Implement Algorithm using JavaScript	15	40	55
Total hours module 11			40	90	130
12	Backend System Design	Analyse System Backend	12	5	17
		Develop System Structure	5	26	31
		Build System Design	5	27	32
Total hours module 12			22	58	80
13	PHP Programming	Apply PHP Fundamentals	15	25	40
		Connect PHP to the Database	7	30	37
		Build a Content Management System (CMS) using PHP	5	20	25
		Build a web app using MVC Framework (LARAVEL)	5	23	28
Total hours module 13			32	98	130
14	Backend Application using Node.js	Develop RESTFUL APIs with Node JS	10	35	45
		Secure Backend Application	5	15	20

		Test Backend Application	5	15	20
		Manage Backend Application	5	10	15
Total hours module 14			25	75	100
15	Windows Server Administration	Manage Server Services	10	25	35
		Manage Users	5	25	30
		Deploy web application	5	20	25
Total hours module 15			20	70	90
16	Industrial Attachment Program	Investigate and secure industrial attachment place.	8	2	10
		Deal with workplace challenges.	7	3	10
		Get briefed on industrial attachment program.	8	2	10
		Develop one’s competencies on the workplace.	7	163	170
Total hours module 16			30	170	200
Total hours for all modules			336	844	1180

3. TRAINING PACKAGE

The training package includes the competencies chart, the flowchart, the modules, the course structure, and the assessment guidelines.

3.1 Course structure

The course structure describes the learning outcomes for . These learning outcomes are the essential skills and knowledge to be acquired. The contents to be covered for each learning outcome are prescriptive. The Learning Activities contain a series of suggestions, usually with several options, that will guide the learner and the trainer.

3.2 Flowchart

The flowchart of sequencing of learning is a schematic representation of the order of acquisition of the competencies. It provides an overall planning of the entire training programme and shows the relationship between the modules. This type of planning is to ensure consistency and progression of learning. For each module, the flowchart shows the learning that is already in place, the learning that is to take in parallel or later. The positions defined will have a decisive impact on all subsequent pedagogical choices. The flowchart of the sequence of learning of the modules of the training programme is presented on the following page.

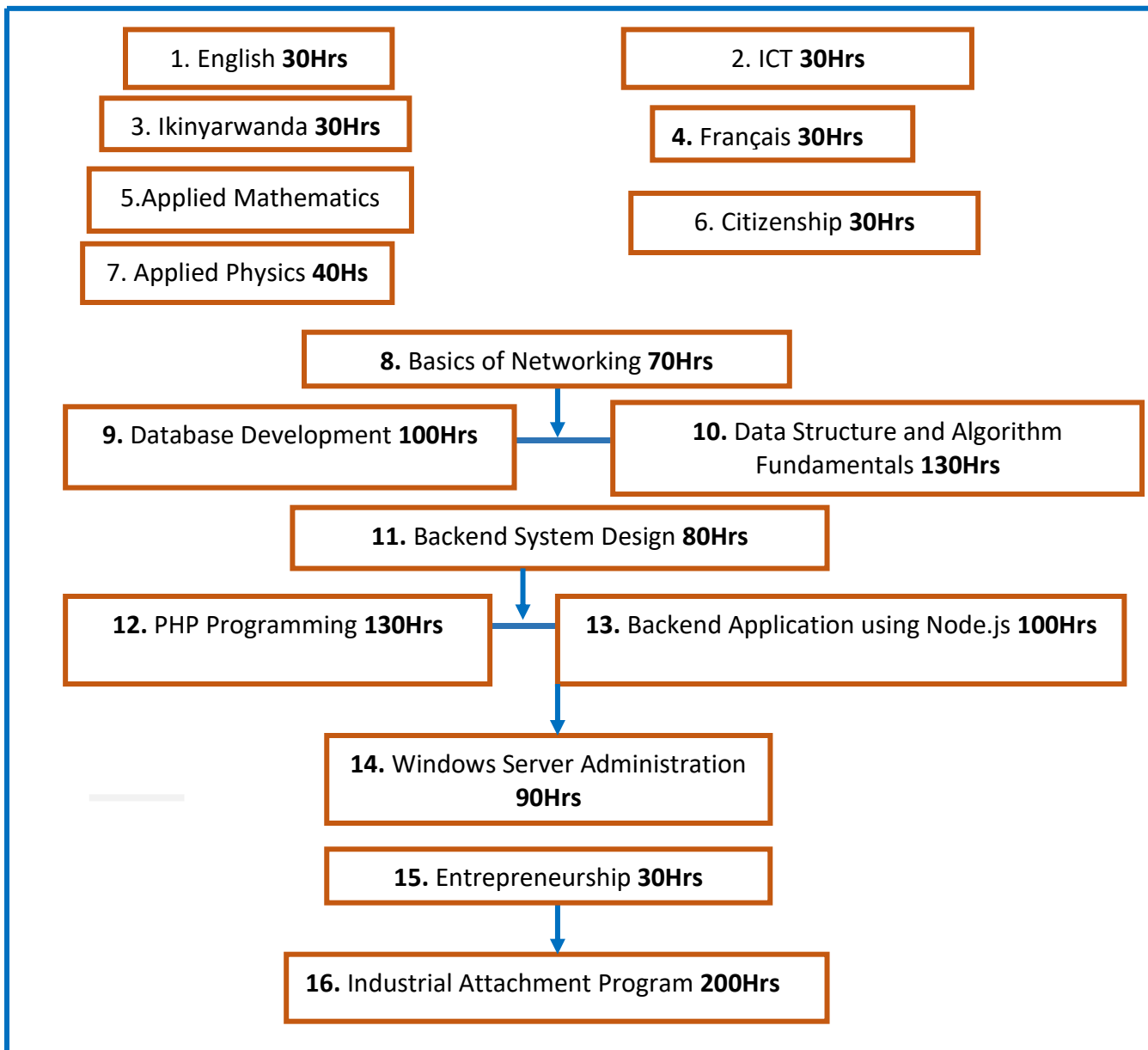


Figure 1 Flowchart

4. ASSESSMENT GUIDELINES

4.1 Assessment Methodology

To assess knowledge, practical, and application skills through a jury system of continuous evaluation that encourages learners to display understanding of the principles in application to set practical tasks and their attendant theory to assess self-learning.

4.2 Portfolio

A portfolio is a collection of learner work representing learner performance. It is a folder (or binder or even a digital collection) containing the learner's work as well as the learner's evaluation of the strengths and weaknesses of the work. Portfolios reflect not only work produced (such as papers and assignments, direct demonstration, indirect demonstration, products, documents), but also it is a record of the activities undertaken over time as part of learner learning. The portfolio is meant to show learner growth, development, and achievements in the education system. It also shows that you have met specific learning goals and requirements. A portfolio is not a project; it is an ongoing process for the formative assessment. The portfolio output (formative assessment) will be considered only as enough for complementary and general modules. Besides, it will serve as a verification tool for each candidate that he/she attended the whole training before he/she undergoes the summative assessment for specific modules

There are two types of assessment (Formative Assessment and Summative/Integrated Assessment). Each assessment has its own rule for passing to be declared competent.

Formative Assessment

This is applied on all types of modules (e.g. Complementary, General and Specific modules)

A trainee to be competent for a Specific module must have at least 70%.

A trainee to be eligible to undertake integrated assessment of specific modules must have at least 50% as passing line for general and complementary modules.

Each trainee should be competent on all formative assessments to be declared competent on that module

All formative assessment should be declared competent before taking the summative/integrated assessment

Summative/Integrated Assessment

All Summative/Integrated assessment should match with the content of the module in the curriculum.

Summative/Integrated Assessment is always practical, giving it as a theoretical type of assessment is not acceptable.

The integrated situation provided in the curriculum is a sample of the assessment to be carried out, the Trainer/Teacher has the role of developing another one referring to the task to be carried out in the integrated situation in accordance to the circumstances inside school, but the integrated situation should stick to the components of a task.

During Summative/Integrated assessment, assessor panel members should be three (3).

The trainee can be declared competent based on the assessment criteria and its respective assessment indicators

The Passing Line for the modules is:

50 % for general and complementary modules

70 % for specific modules

Training delivery		100%	Assessment		Total 100%
Theoretical content		30%	Formative assessment	30%	50%
Practical work:		70%		70%	
Group project and presentation	20%				
Individual project /Work	50%				
		Summative Assessment			50%
Total					100%

Summative assessment is always conducted at the completion of module delivery. It should be practical through an integrated situation for specific modules and in any other forms of assessment for complementary and general modules.

GLOSSARY

Assessment: A process of gathering and judging evidence in order to decide whether a person has attained a standard of performance.

Assessment criteria: Statements which describe performances and place them in context with sufficient precision to allow valid and reliable assessment.

Best practice: Management practices and work processes that lead to outstanding or top-class performance and provide examples for others.

Competency standard: An industry-determined specification of performance which sets out the skills, knowledge and attitudes required to operate effectively in employment. Competency standards are made up of units of competency, which are themselves made up of elements of competency, together with performance criteria, a range of variables, and an evidence guide.

Competency: means the ability to apply knowledge, skills and personal, social and methodological skills in the workplace

or during learning, as well as in personal and professional development. This ability or capacity is acquired through leaning, exposure to the tasks and series of training allowing one to perform specific task autonomously. Reason why in the context of the CBE Framework competencies are described as responsibility and independence.

Competency-based assessment (or CBA): The gathering and judging of evidence in order to decide whether a person has achieved a standard of competency.

Complementary competencies: Set of knowledge, skills and attitudes which are not directly linked to a specific occupation or industry, but which are important for work, education and life in general, such as communication, mathematics, organizational aptitude, and computer literacy, interpersonal and analytical competency.

Core modules: Modules leading to competencies' acquisition that an industry sector has agreed upon as essential for a person to be accepted as competent at a particular level. All modules may be core, but

in many cases competency at a level will involve core modules plus optional or specialization modules. Core competencies are normally those central to work in a particular industry.

Credential: Formal certification issued for successful achievement of a defined set of outcomes, e.g. successful completion of a course in recognition of having achieved particular knowledge, skills or competencies; successful completion of an apprenticeship or traineeship.

Credit: The acknowledgement that a person has satisfied the requirements of a module.

Curriculum: The specifications for a course or subject (module) which describe all the learning experiences a learner undergoes, generally including objectives, content, intended learning outcomes, teaching methodology, recommended or prescribed assessment tasks, assessment exemplars, etc.

Evidence guide: The part of a competency standard which provides a guide to the interpretation and assessment of the unit of competency, including the aspects which need to be emphasized in assessment,

relationships to other units, and the required evidence of competency.

Flexible delivery: A range of approaches to providing education and training, giving learners greater choice of when, where and how they learn. Flexible delivery may involve distance education, mixed-mode delivery, online education, self-paced learning, self-directed learning, etc.

Formal education: Also formal training education or training provided in educational institutions such as schools, universities, colleges, etc. or off the job in a workplace, usually involving direction from a teacher or instructor.

General competencies: competencies correspond to larger operations that go beyond the tasks, but generally contribute to their implementation. These activities require more fundamental learning and are generally common to several tasks and transferable to many work situations.

Generic modules: Modules leading to the attainment of complementary competencies.

Informal education: The acquisition of knowledge and skills through experience, reading, social contact, etc.

Internship: An opportunity for a learner to integrate career related experience by participating in planned, supervised work.

Key competencies: Any of several generic skills or competencies considered essential for people to participate effectively in the workforce. Key competencies apply to work generally, rather than being specific to work in a particular occupation or industry. The following are key areas of competency which were developed into seven key competencies: collecting, analyzing and organizing information; communicating ideas and information; planning and organizing activities; working with others and in teams; using mathematical ideas and techniques; solving problems; and using technology.

Knowledge: means the result of the adoption of information through the learning process. Knowledge is a set of facts, principles, theories and practices related to area of work or study. In CBE context lifelong learning knowledge is described as theoretical and / or factual.

Learning activities: Suggested activities that can be developed during lesson planning and activity preparation. The choice of learning activities must be tailored according to group size, available material resources and communication tools.

Learning hours: Amount of hours required to acquire the competency, including the time allocated to evaluation, which is estimated between 5 and 10% of the total learning time of the competency.

Learning outcomes: Statements that indicate what learners will know or be able to do as a result of a learning activity. Learning outcomes are usually expressed as knowledge, skills, or attitudes.

Module: A unit of training which corresponds to one competency and which can be completed on its own or linked to others.

Occupation: The principal business of one's life.

Performance criteria The part of a competency standard specifying the required level of performance in terms of a set of outcomes which need to be achieved in order to be deemed competent. It describes the

quality requirements of the result obtained in labor performance.

Qualification: means the formal name for the result of a process of assessment and validation, which is obtained when a competent body determines that an individual has achieved learning outcomes to the standards laid down.

Quality assurance: The systems and procedures designed and implemented by an organization to ensure that its products and services are of a consistent standard and are being continuously improved.

Recognition of prior learning (or RPL): The acknowledgement of a person's skills and knowledge acquired through previous training, work or life experience, which may be used to grant status or credit in a subject or module.

Skills: are the ability to apply knowledge and use the principle of "know how" to perform a specific task and to solve the problem. In the context of the CBE Framework, skills are defined as cognitive (involving the use of logical, intuitive and creative thinking), practical (including physical skill and use of methods, materials, devices and

instruments) and social skills (communication and cooperation skills, emotional intelligence and other).

Specific competencies: Competencies that are directly related to the tasks of the occupation in the workplace context. They refer to concrete, practical, and focused aspects

Traineeship: A system of vocational training combining off-the-job training at an approved training provider with on-the-job training and practical work experience. Traineeships generally take one to two years and are now a part of the New Apprenticeships system.

Unit of competency: A component of a competency standard. A unit of competency is a statement of a key function or role in a particular job or occupation. See also element of competency, performance criteria, range of variables.